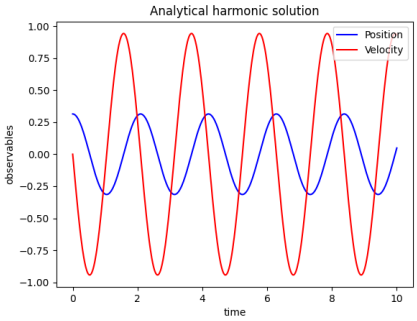
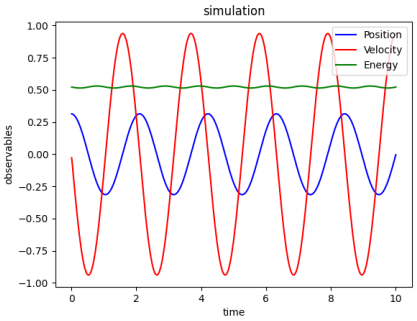
Oscillations and chaos

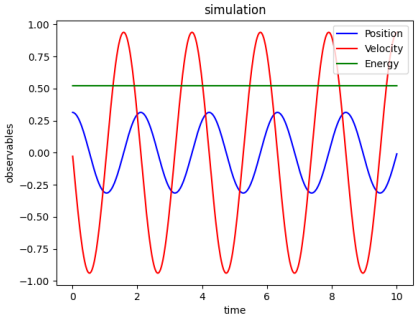
1.1



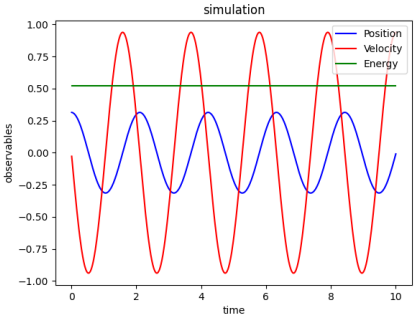
(a) Analytical harmonic solution



(b) Euler-Cromer

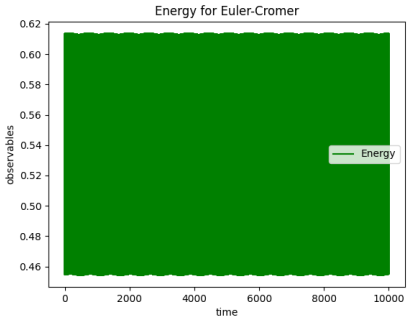


(c) Verlet

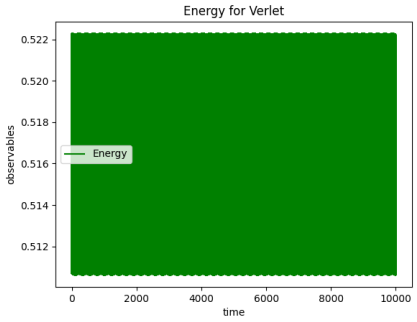


(d) Runge-Kutta 4

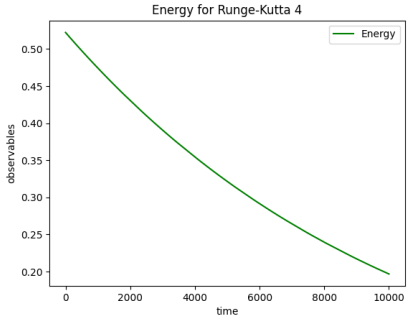
Figure 1: Pendulum simulation with step length 0.01 and initial condition $\theta =$ 0.1*π, $\dot{\theta} = 0$ and $t_{max} = 5$.



(a) Energy for Euler-Cromer



(b) Energy for Verlet

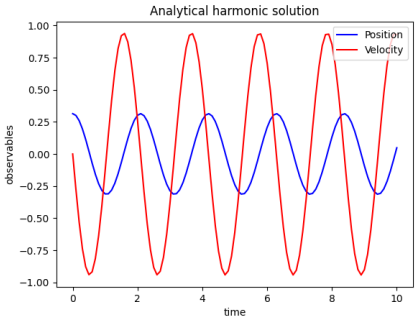


(c) Energy for Runge-Kutta 4

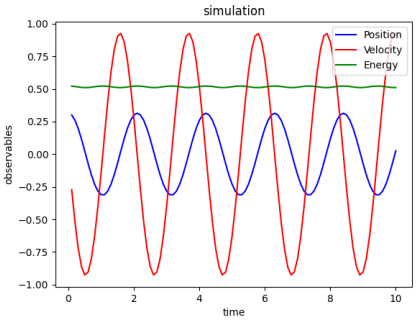
Figure 2: Energies during pendulum simulations. $t_{max} = 10000$ with initial condition $\theta = 0.1{*}\pi$, $\dot{\theta} = 0$ and step length dt = 0.1.

Runge-Kutta 4 is not symplectic according to theory, which can also be seen in figure 2 c). Thus, every iteration is guaranteed to gain a slight cumulative error for the energy, meaning that the numerical solution does not reliably describe the physical system over an extended period of time.

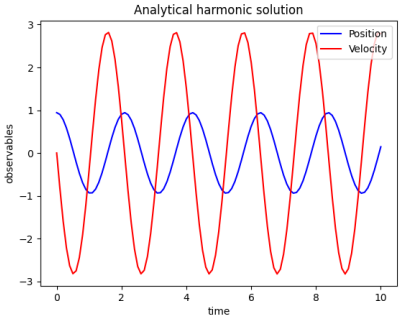
When studying figure 2 a) and b), one can deduce that both Euler-Cromer and Verlet are symplectic. However, the energy of Verlet drifts less than Euler-Cromer, so therefore Verlet is better from an energy perspective.



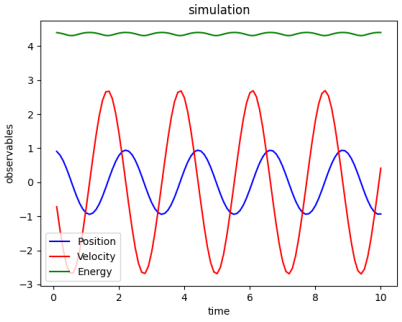
(a) Harmonic, IC: $\theta = 0.1*\pi$



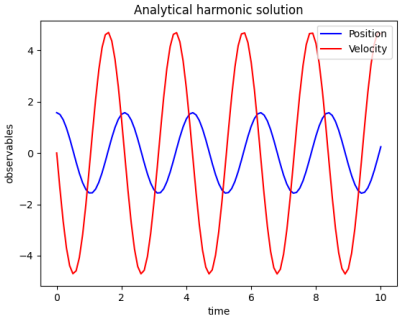
(b) Verlet, IC: $\theta = 0.1*\pi$



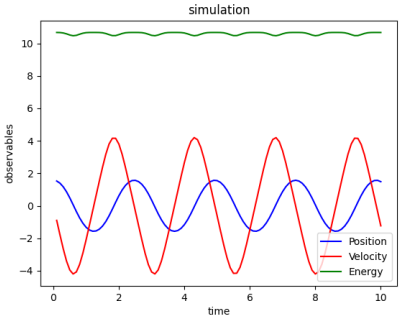
(c) Harmonic, IC: $\theta = 0.3*\pi$



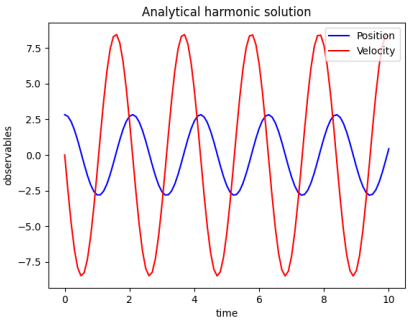
(d) Verlet, IC: $\theta = 0.3*\pi$



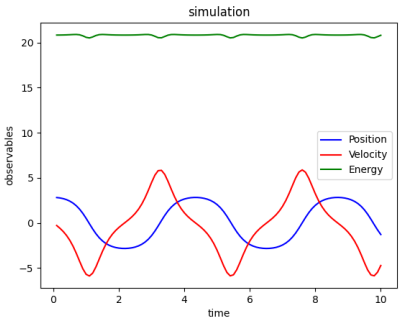
(e) Harmonic, IC: $\theta = 0.5*\pi$



(f) Verlet, IC: $\theta = 0.5*\pi$



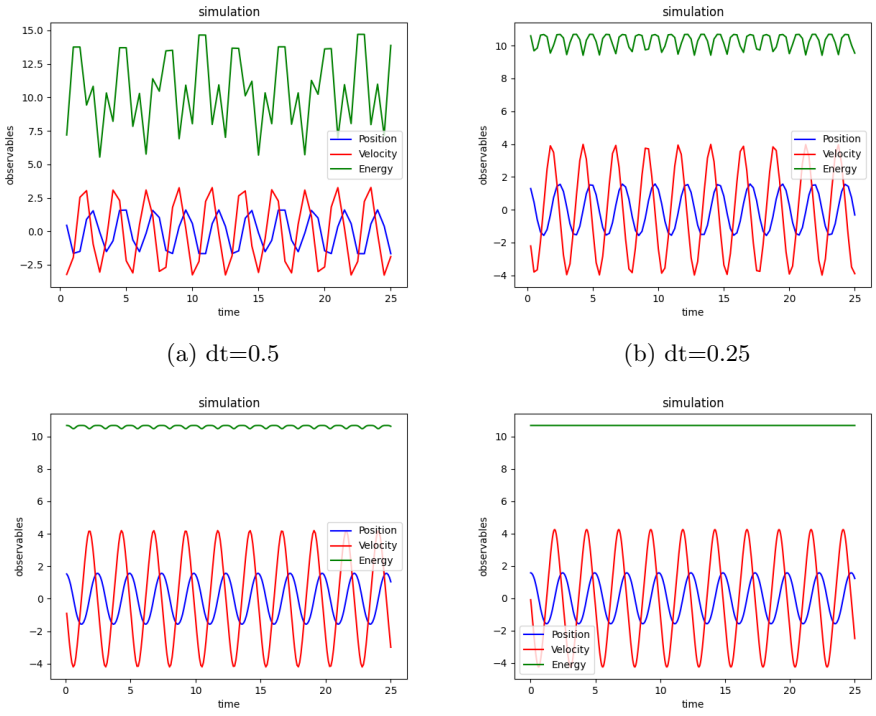
(g) Harmonic, IC: $\theta = 0.9*\pi$



(h) Verlet, IC: $\theta = 0.9*\pi$

Figure 3: Plots of the pendulum solution with Verlet for different initial conditions. $\dot{\theta} = 0$ and step length dt $= 0.1$ for all figures.

As one can see in figure 3, the analytical harmonic solution matches the real pendulum for small angle values, but already starts to visibly deviate with the initial condition $\theta = 0.3^*\pi$. So the small angle approximation works as expected.



(a) dt=0.5

(b) dt=0.25

(c) dt=0.1

(d) dt=0.01

Figure 4: Verlet integration with varying time steps dt and $\theta = 0.5^*\pi$, $\dot{\theta} = 0$ and $t_{max} = 25$

The effect of the step size can be seen in figure 4, with a smaller step yielding a better result. Assuming that $\theta(t) = \bar{\theta}(h) + Kh^p$ (where $\bar{\theta}(h)$ is our numerical solution at t, K is some constant, h is our step size and p is the order of our numerical integrator), one can calculate the time order p as:

$p = \frac{log(E(h)-E(h/2))}{log(2)}$ [1], where $E(h') = |\theta(t) - \bar{\theta}(h)|$ for a given step size h'. Using equation [1], the time step order for Verlet was calculated to p = 2, as seen in figure 5.

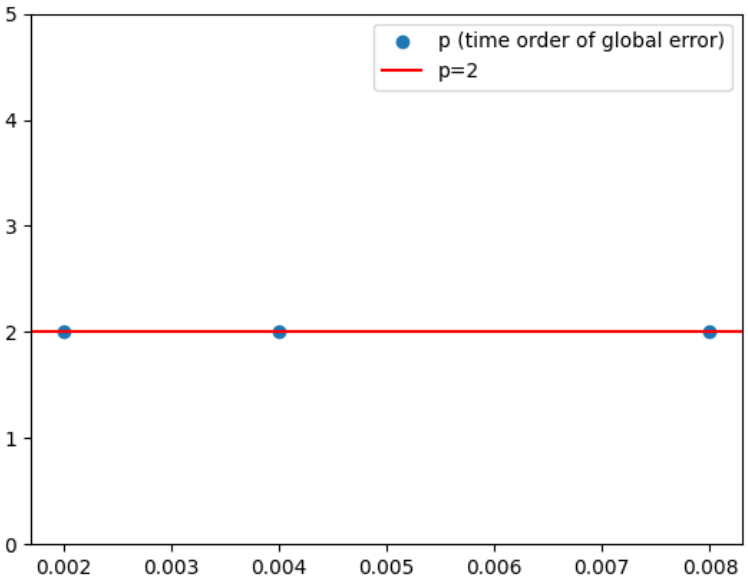


Figure 5: The timeorder was calculated as p = 2, using equation [1] and a RK4 solution with time step $dt_{RK4} = 10^{-6}$ at $t_{max} = 5$ as a replacement for the exact solution. The y axis denotes order, while the x axis denotes the value of h as in equation [1] for the given value of p.

1.2

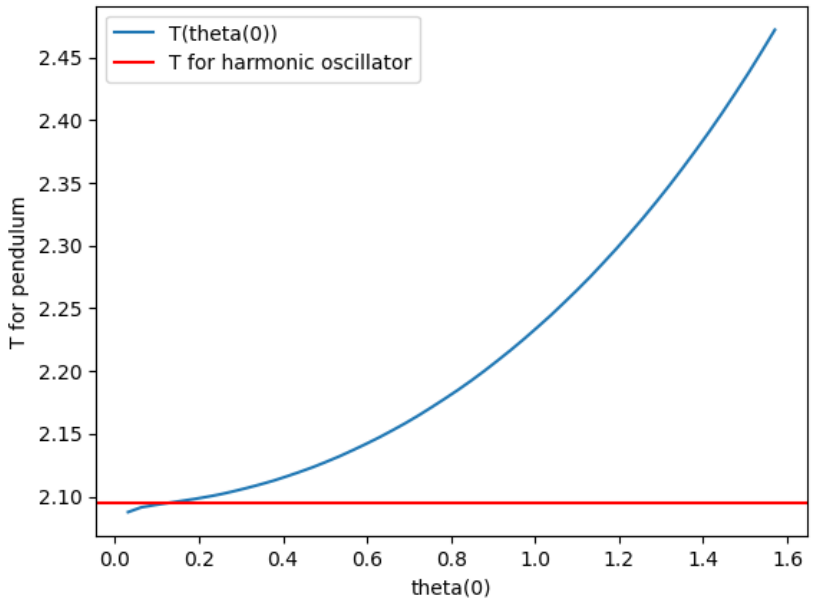


Figure 6: The period time T as a function of the initial condition $\theta(0)$ for both the pendulum (blue) and harmonic oscillator (red). The pendulum period times were calculated with Verlet integration with dt = 0.0001, $\frac{g}{l} = 9$ and m = 1.

As seen in figure 6, the harmonic oscillator and pendulum both have similar period times for small initial conditions due to the small angle approximation. The increased period time for the pendulum makes sense physically, since $|\theta(t)| > |sin(\theta(t))|$, $\forall t$ and $\theta \in [-\pi, \pi]$ (F $= -m\frac{g}{l}\theta$ for harmonic oscillator and F $= -m\frac{g}{l}sin(\theta)$ for the pendulum). Thus the harmonic oscillator applies a larger counteracting force overall, leading to a shortened period time compared to the pendulum. A comparison with the perturbation series can be seen in figure 7.

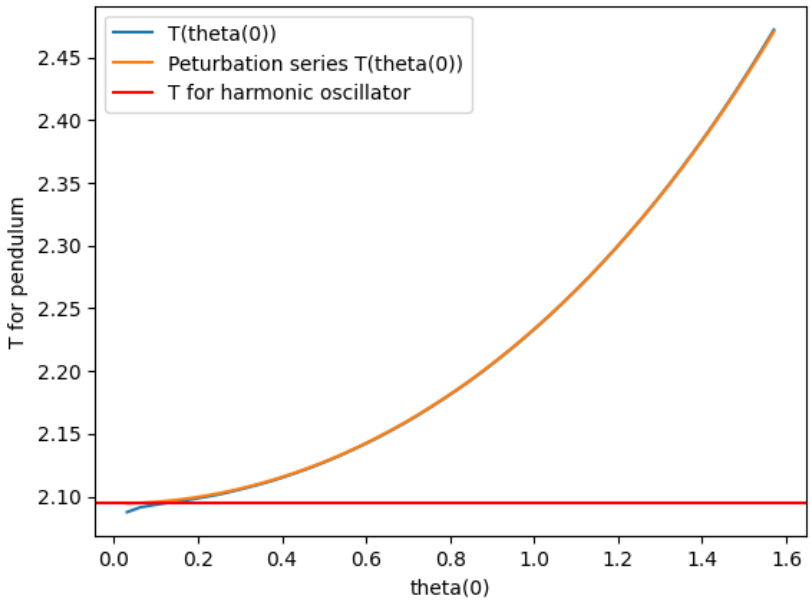
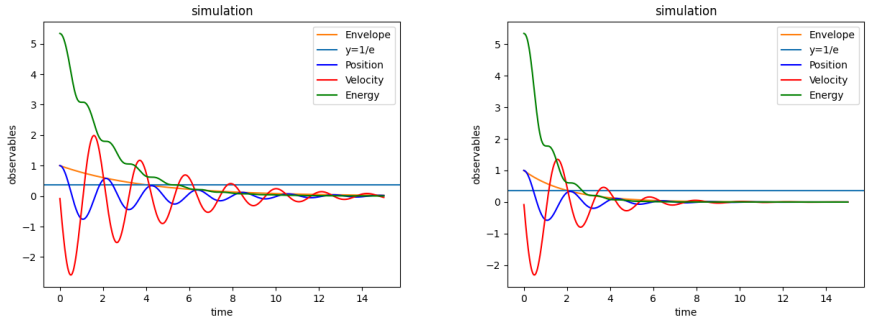


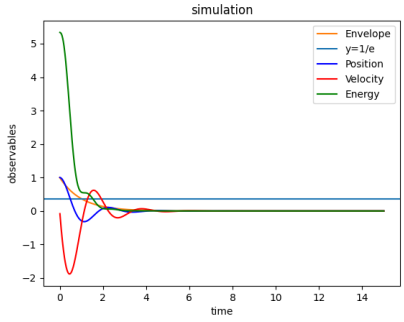
Figure 7: The pendulum period time (blue) and harmonic oscillator period time (red) were calculated in the same way as in figure 6. The perturbation series was evaluated excluding all terms in $\mathcal{O}(\theta(0)^7)$ (orange).

1.3

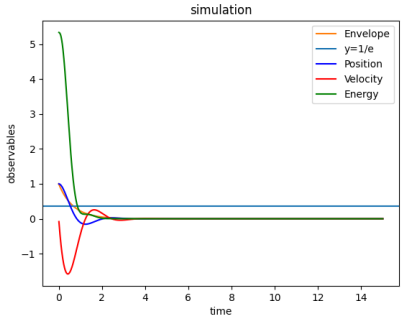


(a) $\gamma = 0.5$

(b) $\gamma = 1$



(c) $\gamma = 2$



(d) $\gamma = 3$

Figure 8: Equation [2] solved in time with $\omega_0 = 3$ and time step dt $= 0.01$.

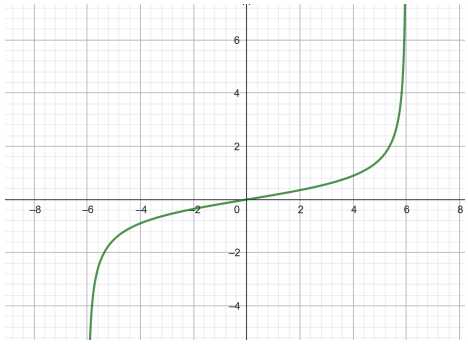


Figure 9: A plot of $f(\gamma) = \frac{\gamma}{2\sqrt{\omega_0^2 - (\frac{\gamma}{2})^2}}$, $\omega_o = 3$.

The solutions of the second order ODE $\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$; $x(0) = 1$, $\dot{x}(0) = 0$ [2], are given by:

- $x(t) = (\frac{1}{2} + \frac{\gamma}{4\sqrt{(\frac{\gamma}{2})^2-\omega_0^2}})e^{(-\frac{\gamma}{2}+\sqrt{(\frac{\gamma}{2})^2-\omega_0^2})t} + (\frac{1}{2} - \frac{\gamma}{4\sqrt{(\frac{\gamma}{2})^2-\omega_0^2}})e^{(-\frac{\gamma}{2}-\sqrt{(\frac{\gamma}{2})^2-\omega_0^2})t}$, $\frac{\gamma}{2} > \omega_0$ [3]
- $x(t) = (1 + \frac{\gamma t}{2})e^{-\frac{\gamma}{2}t}$, $\frac{\gamma}{2} = \omega_0$ [4]
- $x(t) = e^{-\frac{\gamma}{2}t}(cos(\sqrt{\omega_0^2 - (\frac{\gamma}{2})^2}t) + \frac{\gamma}{2\sqrt{\omega_0^2-(\frac{\gamma}{2})^2}}sin(\sqrt{\omega_0^2 - (\frac{\gamma}{2})^2}t))$, $\frac{\gamma}{2} < \omega_0$ [5]

As seen in figure 9, if γ is small, $f(\gamma)$ is also small, meaning that the envelope's phase dependency can be disregarded and thus the envelope can be approximated with $e^{-\frac{\gamma}{2}t}$. Thus we get that: $x(\tau) \sim e^{-\frac{\gamma}{2}\tau} = \frac{1}{e} = e^{-1} \implies \frac{\gamma}{2}\tau = 1 \implies \tau = \frac{2}{\gamma}$. For $\bar{\gamma} = (0.5, 1, 2, 3)$ we get $\bar{\tau} = (4, 2, 1, \frac{2}{3})$, which tracks with the intersection of $\frac{1}{e}$ and the envelope in figure 8. The relationship between τ and γ ($\tau = \tau(\gamma)$ can be seen in figure 10.

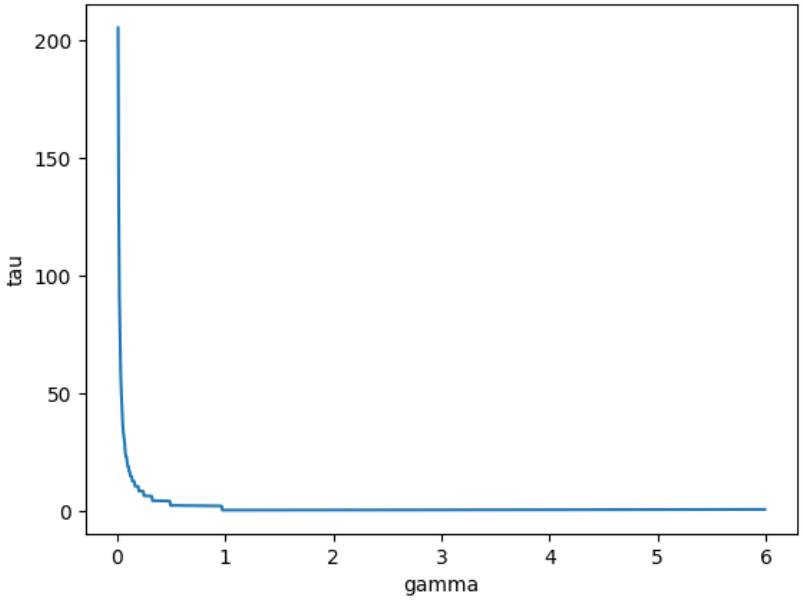


Figure 10: A graph of τ as a function of γ. τ is evaluated at a linspace of γ at 1000 values between 0.01 and 5.99. The value of τ was found through brute force by checking values to see if $x(t) - \frac{1}{e} < tol$, where the *tol* was set to 10^{-2}. The first value was set far to the right, where a manual check of the graph confirmed that $x(t) < \frac{1}{e}$. Then the time t is incremented by dt = 0.001 to the left until the before mentioned condition is met. For every subsequent value of γ, the starting value was set to the previous final value +2.

According to theory, critical damping occurs when the characteristic equation

of equation [2] has a double root, so when $\frac{\gamma}{2} = \omega_0$ according to equation [4]. For $\omega_0 = 3$, we get $\gamma = 6$. The plot for equation [2]'s solution for $\gamma = 6$ is shown in figure 11.

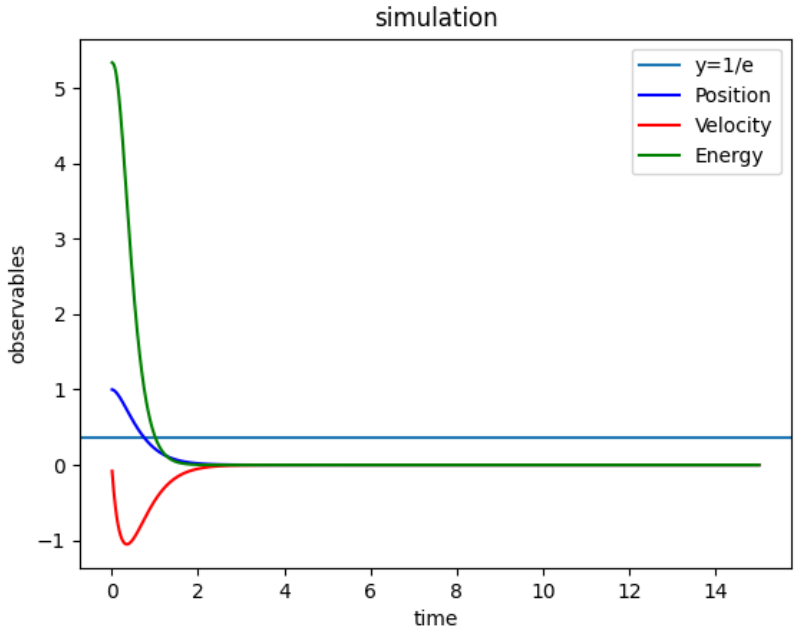


Figure 11: The solution of equation [2], given that $\gamma = \gamma_c = 6$, $\omega_0 = 3$. At the final time, the position is $x = 1.675 \times 10^{-18} > 0$ and since the position is always decreasing, it never becomes negative.

1.4

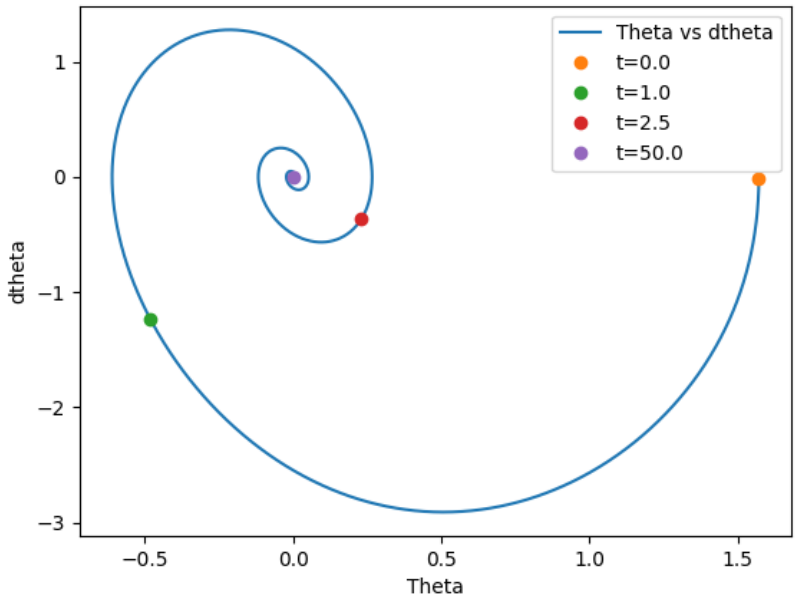


Figure 12: A plot of $\dot{\theta}(t)$ on the vertical axis against $\theta(t)$ on the horizontal axis for $t \in [0, 50]$. The numerical solution for θ and $\dot{\theta}$ was acquired for the pendulum, given $\gamma = 1.5$, $\sqrt{\frac{g}{l}} = 3$, $\theta(0) = \frac{\pi}{2}$, $\dot{\theta}(0) = 0$ and dt $= 0.001$.

The system used for the plot in figure 12 is described by the differential equation: $\ddot{\theta} = -\frac{g}{l} sin(\theta) - \gamma\dot{\theta}$. Using $\dot{\theta} = \xi$, we the following system of equations:

$$\begin{cases} \dot{\theta} &= \xi \ \ [6] \\ \dot{\xi} &= -\frac{g}{l} sin(\theta) - \gamma\xi \ \ [7] \end{cases}$$

From theory of systems of differential equations, with the help of the plotted points at t = {0, 1, 2.5, 50}, we can deduce that the system plotted in figure 12, {[6], [7]}, is stable and if it were linearized near a critical point (say, $(\theta, \dot{\theta}) = (0, 0)$), the eigenvalues would have a negative real part (due to stability) and a negative imaginary part (due to clockwise rotation).

1.5

1.5.1 a)

The steady state behavior of the double pendulum system (initial condition: $|q_{1,2}| < \pi$, p1=0, $E_{1,2,3} = 1, 15, 40$, $t_{max} = 50$, dt $= 0.003$) was first assessed visually by looking at an animation of the simulation.

For E = 1, the following data points were assessed:

$$(q_1, q_2) = \begin{cases} (-0.02577316987841722, -0.28472349560817456) = p_1 \\ (-0.031578679004203115, 0.2780806192467001) = p_2 \\ (0.10648722417768763, 0.3108064174218464) = p_3 \end{cases}$$

Through visual assessment, low energy double pendulum systems seem to have relatively regular movement.

For E = 15, the following data points were assessed:

$$(q_1, q_2) = \begin{cases} (-0.7900806429331356, 0.8073490621677503) = p_4 \\ (0.45811314122630503, -1.4507902034808535) = p_5 \\ (-0.4345072421886303, 0.19516906403575413) = p_6 \end{cases}$$

For E = 15, the system looked more chaotic than when E = 1. Although some aspects of the movement were predictable, like that the lower a arm never does a full revolution, it was difficult to discern any periodic patterns for q1, q2, p1 and p2.

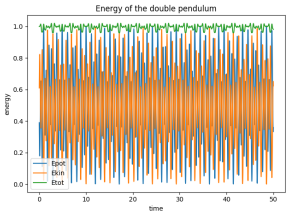
For E = 40, the following data points were assessed:

$$(q_1, q_2) = \begin{cases} (-2.502467983187751, 0.6128931141728367) = p_7 \\ (0.9446494197209796, -0.3041035185385021) = p_8 \\ (1.1058531550558046, 3.061392570207309) = p_9 \end{cases}$$

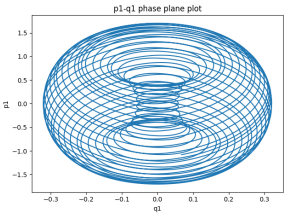
For E = 40, the movement was completely chaotic, with no real pattern for any of the movement.

1.5.2 b)

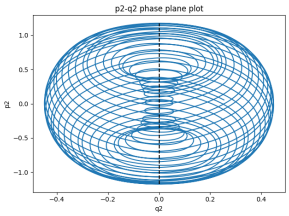
The same datapoints are used in b) as in a) for $E_{1,2,3} = 1, 15, 40$.



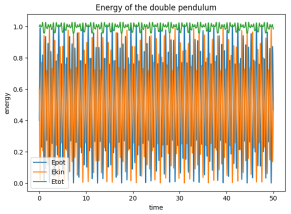
(a) p_1: The energy of the system.



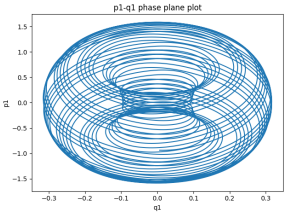
(b) p_1: The phase plane diagram of p1-q1.



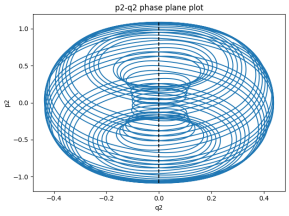
(c) p_1: The phase plane diagram of p2-q2.



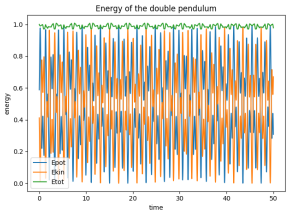
(d) p_2: The energy of the system.



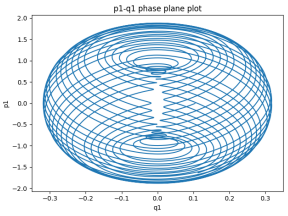
(e) p_2: The phase plane diagram of p1-q1.



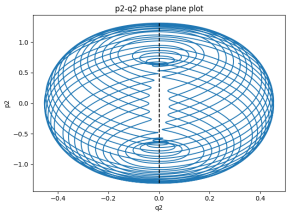
(f) p_2: The phase plane diagram of p2-q2.



(g) p_3: The energy of the system.

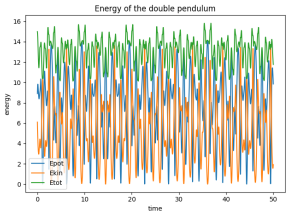


(h) p_3: The phase plane diagram of p1-q1.

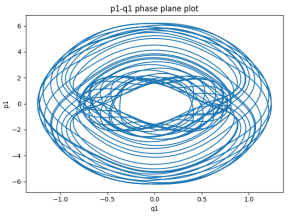


(i) p_3: The phase plane diagram of p2-q2.

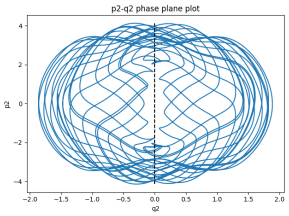
Figure 13: Pendulum simulations for E = 1, dt = 0.003, $t_{max} = 50$.



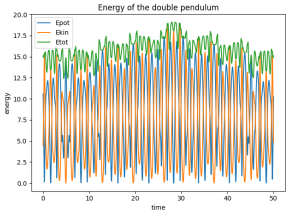
(a) p_4: The energy of the system.



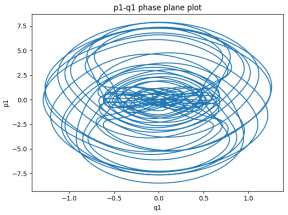
(b) p_4: The phase plane diagram of p1-q1.



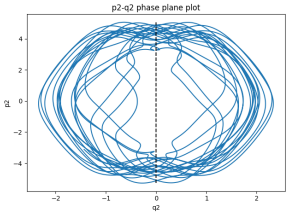
(c) p_4: The phase plane diagram of p2-q2.



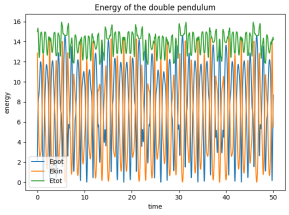
(d) p_5: The energy of the system.



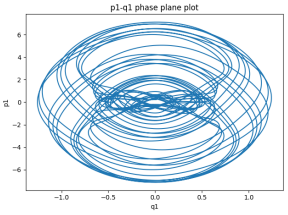
(e) p_5: The phase plane diagram of p1-q1.



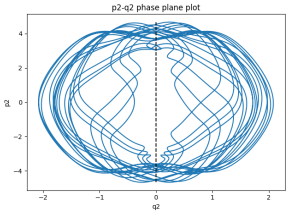
(f) p_5: The phase plane diagram of p2-q2.



(g) p_6: The energy of the system.

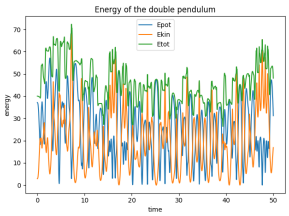


(h) p_6: The phase plane diagram of p1-q1.

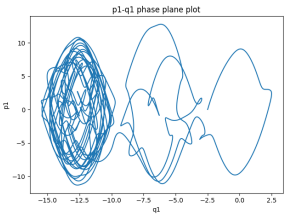


(i) p_6: The phase plane diagram of p2-q2.

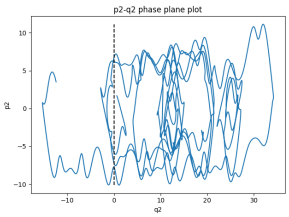
Figure 14: Pendulum simulations for E = 15, dt = 0.003, $t_{max} = 50$.



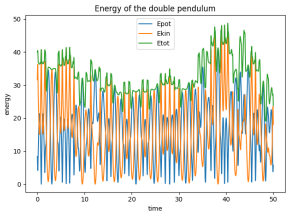
(a) p_7, H=40: The energy of the system.



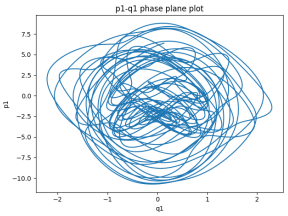
(b) p_7, H=40: The phase plane diagram of p1-q1.



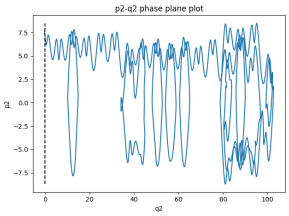
(c) p_7, H=40: The phase plane diagram of p2-q2.



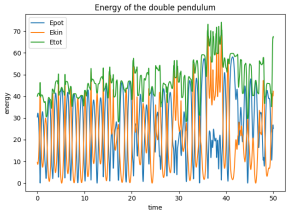
(d) p_8, H=40: The energy of the system.



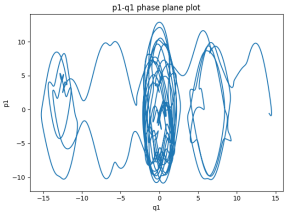
(e) p_8, H=40: The phase plane diagram of p1-q1.



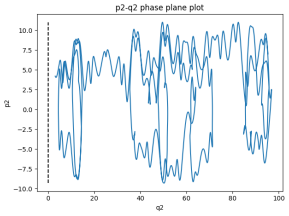
(f) p_8, H=40: The phase plane diagram of p2-q2.



(g) p_9, H=40: The energy of the system.



(h) p_9, H=40: The phase plane diagram of p1-q1.



(i) p_9, H=40: The phase plane diagram of p2-q2.

Figure 15: Pendulum simulations for E = 40, dt = 0.003, t_{max} = 50.

As one can see in the phase space diagrams in figure 13, 14 and 15, the ones for E = 1 are very regular, though not asymptotically stable, while the phase space diagrams for E = 15 are slightly less regular, though they display the same general pattern no matter the initial condition. The energy when E = 1 is also much more stable, so the plot is more reliable than for E = 15, which is slightly less physically realistic.

For E=40, there double pendulum system seems completely chaotic and there is no clear pattern between initial condition and phase space diagram. The energy is also all over the place, so a symplectic method might be better for modelling these kinds of systems.

Since the phase space plot saves the cumulative information of the system's integration until t_{max} into one image, it is significantly easier to tell if the motion is regular or not.

1.5.3 c)

All Poincaré maps below use a tolerance of $tol = 10^{-2}$ for measuring $q_2 = 0$, meaning that we really measure the Poincaré points as: $(q_1, p_1) \mid q_2 < tol$ and $p_2 - p_1 > 0$.

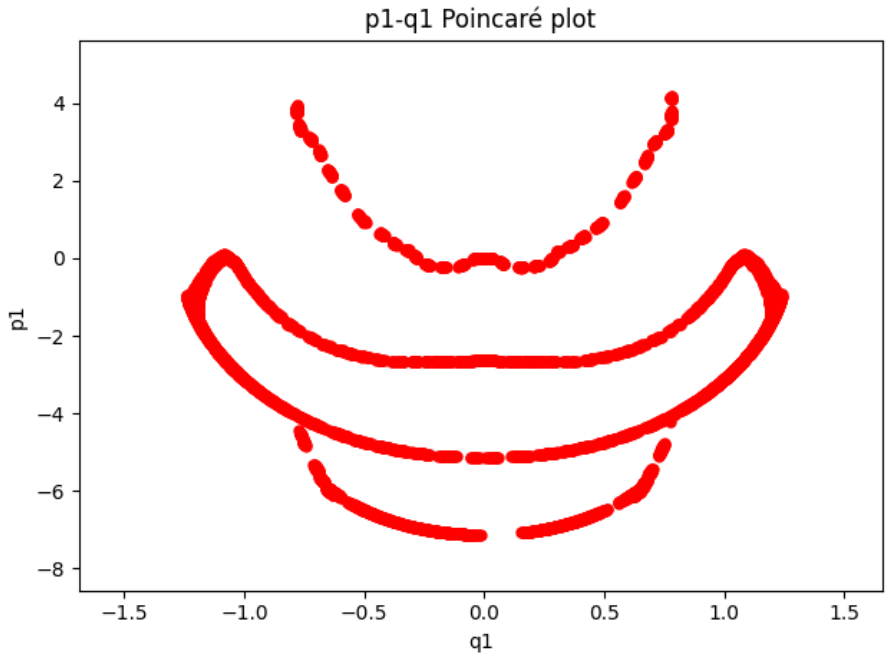
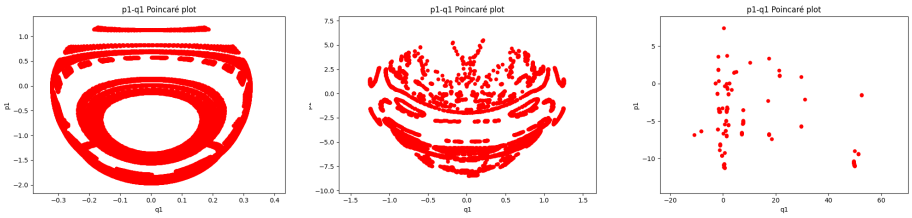


Figure 16: Map meant to replicate "figure 6.13" in the powerpoint for lecture 1. The integration used step length dt = 0.003 and t_{max} = 2000. E = 15 and two initial conditions were simulated: $(q_1, q_2, p_1, p_2) \in \{(0, 0, 0, 3.87); (1.1, 0, 0, 2.77)\}$.

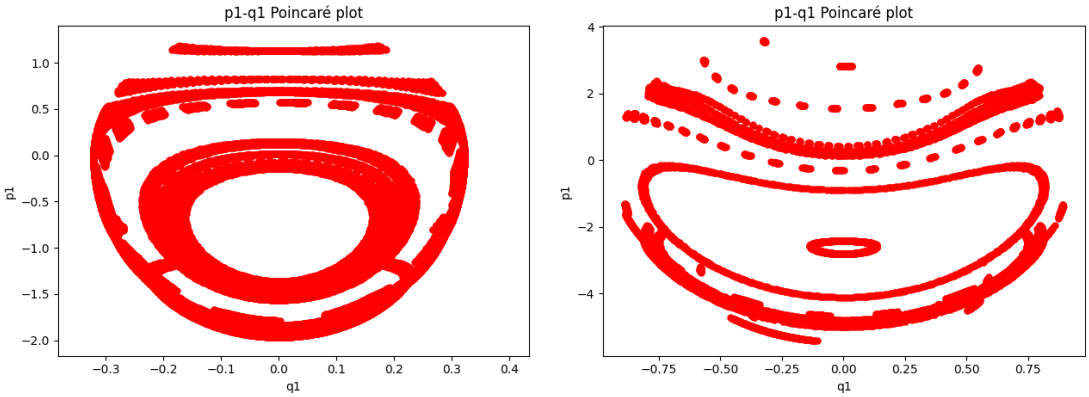


(a) E = 1. (b) E = 15. (c) E = 40.

Figure 17: Each map was generated using Monte Carlo for the initial conditions with n=10 different randomized initial conditions for each map. Each simulated session integrated until $t_{max} = 500$ with dt = 0.003.

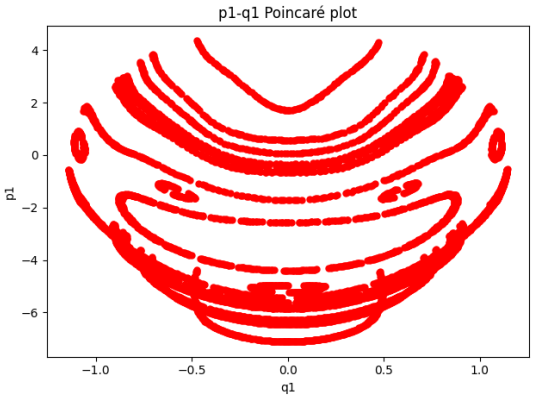
Looking at figure 17, one can see that for E = 1, the pendulum regularly returns to a set of coordinate values (q_1, q_2, p_1), meaning that the movement is, in some sense, regular. For E = 15, we instead see that there are some elements of regularity, but the Poincaré map is much more empty, meaning that there are far more points that are reached, never to be repeated. For E = 40, we see even fewer points, which are also spread out in a seemingly random way.

1.5.4 d)

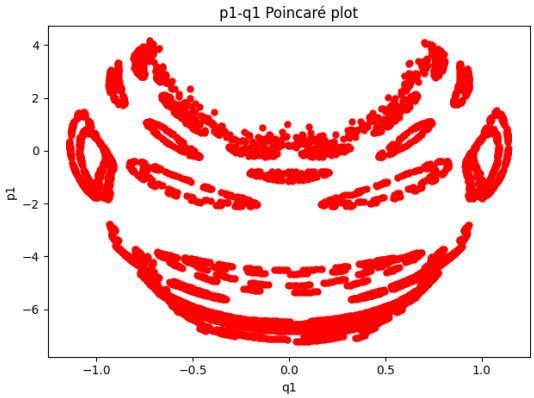


(a) E=1

(b) E=8

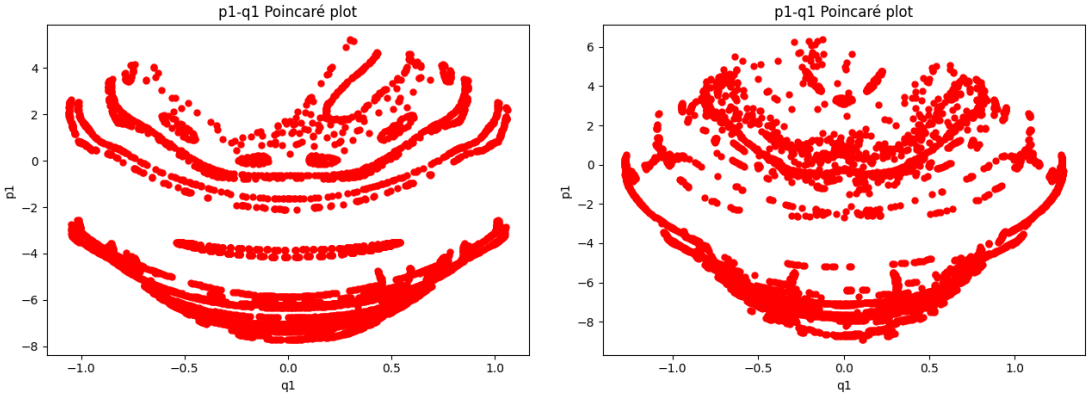


(c) E=12



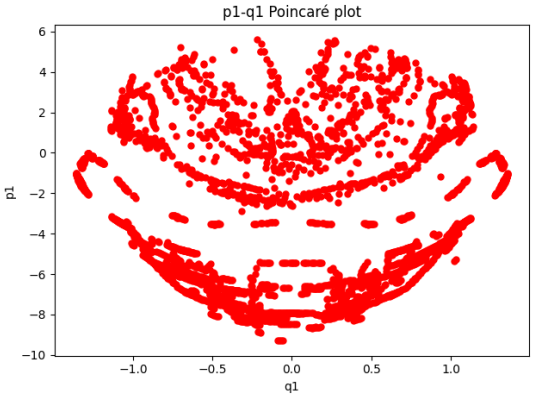
(d) E=13

Figure 18: Poincaré maps for several energies ($E \in 1, 8, 12, 13$) given 10 different random initial conditions per map, integrated until $t_{max} = 500$.

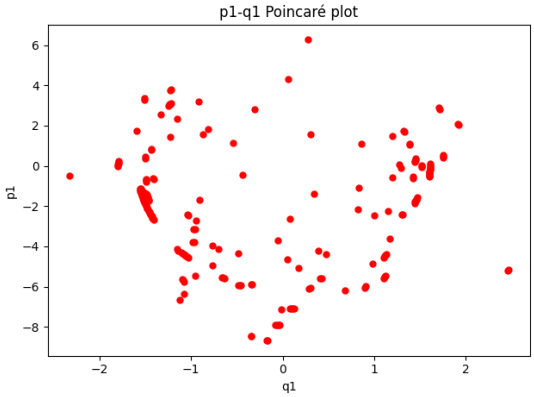


(a) E=14

(b) E=16



(c) E=18



(d) E=25

Figure 19: Poincaré maps for several energies ($E \in 14, 16, 18, 25$) given 10 different random initial conditions per map, integrated until $t_{max} = 500$.

When energy increases, the system becomes more chaotic. In figure 18 and 19, one can observe a pattern of decreasing amounts of points, which are also more erratically placed with a less defined pattern for their placement. My vibes based approach says that the critical energy value is between 12 and 13, since E=12 is pretty smooth, while E=13 looks a bit grainy.